Highlights

AURORA: Cross-Platform HW/SW Co-Design for Energy-Efficient Keccak-256 Blockchain Anchoring

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- Keccak-256 is evaluated on CPU, GPU, Jetson Nano, and FPGA.
- Memory-aware C++ optimization achieves a 4.76× speedup.
- A Zynq-7000 Keccak-256 IP core achieves an 9.17× speedup.
- FPGA execution provides near-deterministic timing behavior.
- GPU execution is inefficient for the tested Keccak-256 workload.

AURORA: Cross-Platform HW/SW Co-Design for Energy-Efficient Keccak-256 Blockchain Anchoring

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ABSTRACT

Embedded blockchain systems are increasingly considered for tamper-evident anchoring in edge and medical environments, where sensitive records should remain off-chain while their integrity is verifiable on-chain. However, the cryptographic preparation stage, particularly Keccak-256 hashing and block-like record construction, can become a performance and energy bottleneck on constrained devices. This paper presents AURORA, an energy-aware on-chain/off-chain architecture that maps cryptographic preparation to heterogeneous embedded platforms while committing only finalized digests and metadata to the blockchain layer. AURORA is evaluated through a cross-platform implementation of the Keccak-256 pipeline on a workstation, an NVIDIA Jetson Nano, and a Xilinx Zybo Zynq-7000 FPGA. Software implementations are optimized from a baseline C++ version using pre-allocated buffers, lookup-table-based hexadecimal encoding, lightweight timestamp generation, and reduced intermediate allocations. A custom Keccak-256 VHDL IP core is also integrated into a Zynq-7000 hardware/software co-design and compared with an ARM-only baseline. Experiments over 10^6 hashes and ten independent runs show that software optimization improves workstation throughput by 4.76×, while the FPGA co-design achieves an 9.17× speedup over ARM-only execution with near-deterministic timing. GPU execution is inefficient for this workload on both tested GPU targets. The architecture is illustrated through a longitudinal medical-data anchoring scenario inspired by Alzheimer's monitoring, without claiming clinical validation.

1. Introduction

Blockchain-based systems provide tamper-evident record management, decentralized auditability, and cryptographic traceability across distributed infrastructures (Tiwari, Ranjan, Biswal et al., 2025; Nakamoto, 2008; Ethereum Foundation, 2025). These properties are particularly attractive in applications where records must remain verifiable over time even when data producers, gateways, or storage providers cannot be fully trusted. Beyond their initial use in cryptocurrency systems, blockchain technologies are increasingly being considered for edge, Internet-of-Things (IoT), and cyber-physical environments in which distributed devices generate data that must be authenticated, timestamped, and later audited (Frikha, Chaabane, Halima et al., 2023). However, deploying blockchain-related operations close to the data source remains challenging because embedded platforms are constrained by limited processing capability, memory availability, timing predictability, and energy budget.

A major bottleneck in such deployments is the cryptographic preparation layer that precedes on-chain anchoring. Before a record can be committed to a blockchain or to a blockchain-backed audit layer, it must typically be serialized, protected, hashed, linked to previous records, and packaged with metadata. In Ethereum-compatible environments, Keccak-256 is a central primitive for producing fixed-size cryptographic digests, and its repeated execution can dominate the cost of block or transaction preparation on constrained hardware. While high-performance processors

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can absorb this cost through clock frequency and memory bandwidth, the same workload can become prohibitive on edge gateways, wearable nodes, or FPGA-based embedded devices. Consequently, the question is no longer only whether blockchain can provide integrity and traceability, but how its cryptographic preparation layer can be executed efficiently, deterministically, and with acceptable energy cost on heterogeneous embedded platforms.

Healthcare-oriented data traceability provides a representative example of this challenge. Medical records, sensor measurements, imaging summaries, and longitudinal monitoring data are often too large and too sensitive to be stored directly on-chain. A more practical approach is therefore to keep the clinical content off-chain under controlled access, while anchoring only cryptographic digests and selected metadata on-chain. This design supports later integrity verification without exposing sensitive patient data. In neurodegenerative-disease monitoring, such as Alzheimer's-related follow-up, the need for traceability is especially clear because clinical interpretation often depends on the integrity and ordering of records collected over long periods, including cognitive assessments, imaging-derived summaries, and wearable or home-sensor observations. Nevertheless, this paper does not propose or validate a clinical Alzheimer's diagnosis system. Instead, Alzheimer's monitoring is used as a motivating scenario for studying the embedded cryptographic and architectural layer required by blockchain-based medical data anchoring.

Existing blockchain-for-healthcare and blockchain-for-IoT studies have mainly focused on data-sharing models, access control, smart contracts, or protocol-level scalability (De Aguiar, Faiçal, Krishnamachari and Ueyama, 2021; Ktari, Frikha, Ben Amor, Louraidh, Elmannai and Hamdi, 2022; Ghadi, Mazhar, Shahzad et al., 2024; Dorri, Kanhere and Jurdak, 2019). In parallel, several embedded blockchain works have investigated lightweight consensus mechanisms, off-chain storage, and hardware acceleration for selected cryptographic or consensus components (Ktari, Frikha, Hamdi and Hamam, 2024b). However, fewer studies provide a controlled cross-platform analysis of the Keccak-256 preparation pipeline across software-optimized processors, embedded GPU execution, and FPGA-based hardware/software co-design. Such evidence is important for engineering decisions in medical edge infrastructures, where the appropriate deployment target may differ depending on whether the node is a hospital server, a home gateway, or a low-power acquisition device.

This work addresses this gap by proposing AURORA, an embedded on-chain/off-chain architecture for blockchain-based medical data anchoring. The architecture separates the off-chain cryptographic preparation layer from the network-facing anchoring layer. Sensitive records remain off-chain, while only finalized digests and associated metadata are intended to be committed to the blockchain layer. This separation has two engineering advantages. First, it limits the amount of sensitive information exposed to the network-facing environment. Second, it makes the cryptographic preparation stage an isolated optimization target that can be mapped to different hardware substrates, including optimized CPU software, CUDA-capable embedded platforms, and custom FPGA logic.

To evaluate this design, the Keccak-256 hashing pipeline is implemented and tested on three representative hardware platforms. A high-performance workstation provides a reference for software optimization and desktop GPU behavior. An NVIDIA Jetson Nano (NVIDIA Corporation, 2025), combining an ARM processor with a CUDA-capable embedded GPU, represents an edge gateway deployed near the data source. A Xilinx Zybo Zynq-7000 FPGA (AMD (Xilinx), 2025) hosts a custom Keccak-256 VHDL IP core integrated through a hardware/software co-design, representing low-power embedded nodes where timing determinism and energy efficiency are critical. Across these platforms, the study compares baseline software, memory-optimized software, GPU execution, and FPGA-based acceleration using a common workload of 10^6 hashes and ten independent runs per configuration.

The main contributions of this paper are as follows:

- We propose AURORA, an embedded on-chain/off-chain architecture for blockchain-based medical data anchoring, in which sensitive records remain off-chain while cryptographic digests and metadata are prepared for on-chain auditability.
- We design and integrate a custom Keccak-256 VHDL IP core within a Zynq-7000 hardware/software co-design framework and evaluate its acceleration benefit against an ARM-only embedded baseline.
- We provide a cross-platform evaluation of the Keccak-256 cryptographic preparation pipeline on a workstation, an NVIDIA Jetson Nano, and an Xilinx Zybo Zynq-7000 FPGA using 10^6 hashes and ten independent runs per configuration.
- We compare baseline software, memory-optimized software, CUDA execution, and FPGA-based acceleration in terms of execution time, throughput, execution-time variability, and power-normalized efficiency.

- We discuss the applicability of the proposed architecture to longitudinal medical-data anchoring, using Alzheimer's monitoring as a representative scenario, while explicitly distinguishing platform-level validation from clinical validation.

The remainder of the paper is organized as follows. Section 2 reviews blockchain consensus mechanisms, embedded blockchain implementations, hardware acceleration, and hybrid on-chain/off-chain architectures, with emphasis on their relevance to medical data traceability. Section 3 presents the proposed AURORA architecture, its off-chain cryptographic preparation flow, the hardware/software implementation, and the representative medical anchoring scenario. Section 4 describes the experimental methodology and reports the cross-platform results. Section 5 discusses the findings, compares the proposed approach with prior work, and analyzes the main limitations and threats to validity. Section 6 concludes the paper and outlines future research directions.

2. Related Work and Background

This section positions the proposed AURORA architecture with respect to prior work on blockchain consensus mechanisms, embedded blockchain implementations, hardware acceleration, and hybrid on-chain/off-chain data anchoring. The objective is not to provide a general survey of blockchain applications, but to identify the technical gap addressed in this paper: the lack of controlled cross-platform evidence on the execution cost, variability, and power-normalized efficiency of the Keccak-256 cryptographic preparation layer on heterogeneous embedded hardware. Particular attention is therefore given to works involving FPGA acceleration, lightweight blockchain execution, and off-chain anchoring for data-intensive and medical edge environments.

2.1. Blockchain consensus and cryptographic preparation in constrained environments

Consensus mechanisms define how distributed blockchain nodes agree on the validity and ordering of records. Their design affects security, scalability, latency, energy consumption, and the degree of decentralization of the system (De Aguiar et al., 2021). Proof-based mechanisms, such as Proof-of-Work (PoW), provide strong tamper-resistance through computationally expensive nonce search, but their energy cost and limited throughput make them difficult to deploy on constrained embedded platforms. Proof-of-Stake (PoS) and related variants reduce the computational burden by replacing intensive puzzle solving with validator selection based on stake or authority assumptions, but they introduce different trust and governance constraints (Zhou, Diro, Saini, Kaisar and Hiep, 2024).

Voting-based mechanisms, such as PBFT, Raft, and related permissioned-chain protocols, are often more suitable for enterprise, IoT, and medical-edge settings because participants are known or partially trusted. These protocols can provide lower latency and higher throughput than PoW, although their communication overhead may increase rapidly with the number of validators (Xu, Wang and Jia, 2023). Alternative mechanisms, including Proof-of-Authority (PoA), hybrid consensus schemes, directed acyclic graph structures, sharding, and Layer-2 approaches, further illustrate the diversity of design choices available for reducing network-level cost and improving scalability (Guo and Yu, 2022).

However, consensus selection alone does not remove the cost of the cryptographic preparation stage that precedes anchoring. Regardless of whether the final blockchain layer uses PoW, PoS, PoA, or a permissioned consensus protocol, records must still be serialized, hashed, linked to previous records, and packaged with metadata before they can be committed or verified. In Ethereum-compatible environments, Keccak-256 remains a central hashing primitive. For embedded deployments, repeated Keccak-256 execution can therefore become a bottleneck in terms of execution time, timing variability, and energy cost. This motivates the present work, which does not propose a new consensus protocol, but focuses on accelerating and characterizing the off-chain Keccak-256 preparation layer required by blockchain-based data anchoring.

2.2. Blockchain-based traceability for medical and IoT data

Blockchain has been widely investigated for applications requiring tamper-evident traceability, including finance, supply chains, Industry 4.0, governance, education, healthcare, and IoT infrastructures (Kyriazis, 2021; Abrar and Sheikh, 2024; Ren, Wan and Gan, 2021; Talha, Frikha, Ktari and Hamam, 2023; Ktari, Affes, Frikha, Hamdi and Hamam, 2024a; Roumeliotis, Dasygenis, Lazaridis and Dossis, 2024; Wang, Yu and Zhou, 2025; Kostopoulos, Stamatiou, Halkiopoulos and Antonopoulou, 2025; Kudva, Badsha, Sengupta, Khalil and Zomaya, 2021; Gao, Pan, Liu, Lin, Chen and Wen, 2021; Chaabane, Ktari, Frikha and Hamam, 2022). In the context of the present paper, the most relevant application family is medical and IoT data traceability, where the key challenge is not only to store records securely, but also to preserve their provenance, integrity, and chronological ordering over long periods.

Healthcare-oriented blockchain studies commonly emphasize secure data sharing, patient-centric access control, interoperability across institutions, and tamper-evident audit trails (De Aguiar et al., 2021). IoMT platforms have also shown that blockchain can support remote monitoring and e-health infrastructures by anchoring sensor-derived records on-chain while keeping raw data off-chain (Ktari et al., 2022). Broader reviews of blockchain for the Internet of Medical Things confirm that integrity protection and traceability of device-generated data remain central requirements, especially for continuous monitoring, rehabilitation, pandemic response, and distributed medical data exchange (Nguyen, Ding, Pathirana and Seneviratne, 2021; Ghadi et al., 2024).

Medical imaging and AI-assisted diagnosis introduce additional traceability requirements. Blockchain has been proposed to document the provenance of radiological data, support dynamic patient consent, and track the use of imaging data in diagnostic or learning pipelines (Kotter, Marti-Bonmati, Brady and Desouza, 2021). Related work on reversible data hiding and encrypted medical images combines confidentiality mechanisms with on-chain hash anchoring to improve traceability of imaging archives (Horng, Chang, Li, Lee and Hwang, 2021). At the transaction level, blockchain-based smart-contract frameworks combined with edge devices and mobile-edge computing have also been proposed for secure health-information exchange among patients, clinicians, and administrative stakeholders (Ahmed et al., 2024). More recent work on blockchain-assisted federated learning further underlines the importance of on-chain anchoring for cross-institutional medical data processing (Zhang et al., 2026).

These studies demonstrate the relevance of blockchain for medical data traceability, but many of them remain focused on data governance, sharing models, or high-level system architecture. Less attention is given to the hardware-level cost of preparing the cryptographic evidence that such systems rely on. This cost becomes important when the data source is an embedded or edge device, such as a wearable sensor, a home gateway, or a clinic-side acquisition node. In longitudinal neurodegenerative monitoring, including Alzheimer's-related follow-up, the integrity of the sequence of records is particularly important because clinical interpretation may depend on the ordering and completeness of cognitive assessments, imaging-derived summaries, and sensor-based observations. Existing Alzheimer's-oriented blockchain studies often emphasize AI-based screening, federated learning, or high-level IoT integration (Karaman et al., 2024; Kumar et al., 2024; Singh, Tatiya, Shrivastava, Verma, Srivastava and Rana, 2022). In contrast, the present paper focuses on the embedded cryptographic preparation layer that would be required before such records can be anchored and later verified.

2.3. Embedded blockchain implementation and hardware acceleration

Deploying blockchain-related operations on embedded devices requires careful management of computation, memory, latency, and power. Previous studies have explored several strategies, including lightweight clients, local or federated chains, off-chain processing, and hardware acceleration of selected cryptographic or consensus components. FPGA-based acceleration is particularly relevant because it can provide deterministic execution, low latency, and favorable energy efficiency for bit-level cryptographic operations. Prior work in embedded signal-processing and security-oriented systems has already shown that FPGA-based hardware acceleration can substantially improve computational efficiency and responsiveness (Karmani et al., 2009). Lightweight embedded blockchain execution has also been investigated on low-cost platforms; for example, Allouche, Frikha, Mitrea, Memmi and Chaabane (2021) implemented an off-chain blockchain-based document tracking system on Raspberry Pi hardware.

2.3.1. FPGA-based blockchain acceleration

Sakakibara, Nakamura and Matsutani (2017) addressed blockchain scalability for IoT Simplified Payment Verification (SPV) nodes that cannot store the full blockchain. Their system uses an FPGA-based key-value store cache on a Network Interface Card to store Merkle paths of recent transactions. On cache hits, the FPGA can answer SPV requests without involving a full node, reducing server load and improving throughput. The reported throughput gain ranges from 6.73× to 7.45×, although DRAM latency remains a limiting factor.

Ktari et al. (2024b) investigated FPGA-based acceleration for blockchain consensus processing. Their modular architecture uses customizable hash cores and protocol logic to improve throughput, latency, and energy efficiency compared with software implementations. This work confirms the potential of FPGA technology for blockchain-related acceleration, but it remains primarily focused on consensus components rather than on a broader cross-platform evaluation of the cryptographic preparation pipeline.

Frikha, Chaabane, Aouinti, Cheikhrouhou, Ben Amor and Kerrouche (2021) implemented consensus-related processing using parallel FPGA IP cores. Their approach demonstrates the efficiency of hardware-based consensus execution, but part of the block creation and preparation flow remains outside the protected hardware path. This leaves an open question concerning the integration of block preparation, hashing, and anchoring-related operations within a unified off-chain pipeline. AURORA addresses this point by focusing on the Keccak-256 hashing and block-preparation layer and by comparing optimized software, embedded GPU execution, and FPGA-based HW/SW co-design under a common workload.

2.3.2. Embedded on-chain/off-chain architectures

Hybrid on-chain/off-chain architectures are commonly adopted when the data to be protected is too large, frequent, sensitive, or costly to store directly on-chain. In such systems, raw data remain off-chain, while the blockchain stores cryptographic commitments, hashes, metadata, or verification paths. This approach reduces storage cost and improves scalability while preserving tamper-evident verification through on-chain anchors (Gong and Min, 2023; Eren, Karaduman and Gençoğlu, 2025).

Several design patterns support this separation. Commit-reveal schemes, witnesses, dispute games, and authenticated data structures can be used to preserve integrity and verifiability while reducing on-chain computation and storage (Eberhardt and Tai, 2017; Wöhrer and Zdun, 2021). Thin clients and stateless clients, combined with fraud proofs or data-availability sampling, allow constrained nodes to verify blockchain data without storing the full chain (Sel, Jacobsen et al., 2018). Merkle trees, Verkle trees, and Merkle Patricia Tries provide authenticated query mechanisms for off-chain storage while maintaining an on-chain cryptographic root (Wang, Zhang, Yang, Wang, Zhang and Yuan, 2022; Liu, Peng, Xu, Jiang, Wu and Wang, 2024).

For IoT and embedded systems, layered or federated blockchain architectures can reduce latency and limit dependence on permanent connectivity with a main chain. Local subchains or edge-level validation layers can process data close to the source, while higher-security chains are used for aggregation and anchoring (Lunardi, Alharby, Nunes, Zorzo, Dong and Middleton, 2020). Governance and security models must then cover both environments, since vulnerabilities may arise not only on-chain, but also in off-chain storage, gateways, or oracle-like components (Ahmadjee, Mera-Gómez, Bahsoon and Kazman, 2022).

In the context of this paper, the on-chain/off-chain separation is used for a narrower and more implementation-oriented purpose. Sensitive medical records remain off-chain, while the Keccak-256 hashing and block-preparation stage is isolated as a hardware-optimizable computation. The resulting digests and metadata are intended to be anchored on-chain. This design does not claim to solve all governance or clinical deployment issues; rather, it characterizes the execution layer required to support secure and energy-aware anchoring on embedded hardware.

Table 1 compares representative embedded, hybrid, and hardware-accelerated blockchain approaches using feature-level criteria. The comparison highlights whether each work addresses embedded deployment, hardware acceleration, Keccak-256 preparation, hybrid anchoring, cross-platform validation, and medical-data anchoring.

Table 1: Feature-based positioning of related blockchain approaches

Work	Focus	Edge	HW accel.	Keccak prep.	Hybrid	Cross-platform	Medical	Main limitation
Sakakibara et al. (2017) (Sakakibara et al., 2017)	SPV caching	(✓)	(✓)	(×)	(×)	(×)	(×)	No Keccak preparation study.
Allouche et al. (2021) (Allouche et al., 2021)	Document tracking	(✓)	(×)	(×)	(✓)	(×)	(×)	No hardware acceleration.
Frikha et al. (2021) (Frikha et al., 2021)	Embedded consensus	(✓)	(✓)	(△)	(×)	(×)	(×)	Single FPGA-oriented setup.
Ktari et al. (2024) (Ktari et al., 2024b)	Consensus acceleration	(✓)	(✓)	(△)	(×)	(×)	(×)	Consensus-level focus.
Gong and Min (2023) (Gong and Min, 2023)	Storage scalability	(×)	(×)	(×)	(✓)	(×)	(×)	No embedded evaluation.
Eren and Gençoğlu (2025) (Eren et al., 2025)	Hybrid storage	(×)	(×)	(×)	(✓)	(×)	(×)	Review-only study.
Wöhrer and Zdun (2021) (Wöhrer and Zdun, 2021)	Architecture decisions	(×)	(×)	(×)	(△)	(×)	(×)	No performance results.
Wang et al. (2022) (Wang et al., 2022)	Query authentication	(×)	(×)	(×)	(✓)	(×)	(×)	No hashing acceleration.
Liu et al. (2024) (Liu et al., 2024)	Query verification	(×)	(×)	(×)	(✓)	(×)	(×)	No embedded execution.
Lunardi et al. (2020) (Lunardi et al., 2020)	IoT consensus	(✓)	(×)	(×)	(△)	(×)	(×)	No HW/SW co-design.
Ahmadjee et al. (2022) (Ahmadjee et al., 2022)	Threat modeling	(×)	(×)	(×)	(△)	(×)	(×)	No implementation results.
This work	**Keccak anchoring**	(✓)	(✓)	(✓)	(✓)	(✓)	(✓)	**Unified embedded HW/SW evaluation.**

Legend: (✓) = addressed; (△) = partially addressed; (×) = not addressed. *Edge*: embedded or edge deployment; *HW accel.*: hardware acceleration; *Keccak prep.*: Keccak-256 preparation layer; *Hybrid*: hybrid on-chain/off-chain design; *SPV*: Simplified Payment Verification; *HW/SW*: Hardware/Software.

Table 1 shows that prior work mainly focuses on storage scalability, query verification, consensus optimization, architectural design, or isolated FPGA acceleration. In contrast, this work combines embedded execution, Keccak-256 preparation, FPGA-based HW/SW co-design, cross-platform evaluation, and medical data anchoring within a unified experimental framework.

2.4. Research gap and positioning

The reviewed literature reveals three main limitations. First, many blockchain-for-healthcare and blockchain-for-IoT studies focus on data sharing, access control, smart contracts, or protocol-level scalability, while providing limited evidence on the hardware-level cost of cryptographic preparation on embedded devices. Second, FPGA-based blockchain studies demonstrate the value of hardware acceleration, but often target specific consensus components or isolated primitives rather than comparing software, GPU, and FPGA implementations under a unified experimental methodology. Third, hybrid on-chain/off-chain architectures are widely recognized as useful for scalable and privacy-preserving data anchoring, but their deployment on heterogeneous edge hardware is rarely evaluated in terms of execution time, throughput, execution-time variability, and power-normalized efficiency.

AURORA addresses this gap by combining: (i) an embedded on-chain/off-chain architecture in which sensitive medical records remain off-chain while cryptographic digests and metadata are prepared for blockchain anchoring; (ii) a custom Keccak-256 hardware accelerator integrated within a Zynq-7000 HW/SW co-design framework; and (iii) a cross-platform evaluation methodology spanning a workstation, an NVIDIA Jetson Nano edge platform, and a Xilinx Zybo Zynq-7000 FPGA. The evaluation is platform-centric rather than clinically validated. Alzheimer's-related longitudinal monitoring is therefore used as a representative medical anchoring scenario, not as a claimed diagnostic or clinical validation contribution.

3. Proposed Approach

3.1. Overview of the AURORA architecture

Figure 1 presents the proposed AURORA architecture for embedded blockchain-based medical data anchoring. The architecture is organized into three functional layers: an acquisition layer, an off-chain cryptographic preparation layer, and an on-chain anchoring layer. The main design principle is to keep sensitive records outside the blockchain while committing only cryptographic digests and selected metadata to an immutable audit layer. **Acquisition layer.** The acquisition layer represents the data-producing side of the system. In a medical edge scenario, records may originate from imaging devices, cognitive-assessment tools, wearable sensors, home IoT gateways, or clinic-side acquisition systems. These sources may generate heterogeneous records with different sizes, frequencies, and privacy requirements. The proposed architecture does not require these raw records to be stored on-chain. Instead, they remain under controlled off-chain storage, where access control, pseudonymization, and domain-specific preprocessing can be applied according to the deployment policy.

Off-chain cryptographic preparation layer. The off-chain layer performs the computationally intensive preparation steps required before blockchain anchoring. A record is first formatted and associated with metadata such as timestamp, device identifier, record type, and pseudonymous subject identifier. The resulting payload is then processed by the Keccak-256 hashing pipeline, which produces a fixed-length digest used as an integrity anchor. The digest and metadata are then packaged into a block or transaction-ready structure. This layer is the primary optimization target of the present work, because repeated Keccak-256 execution and block preparation can become a bottleneck on constrained devices.

On-chain anchoring layer. The on-chain layer stores only the finalized digest and associated metadata required for later verification. The raw medical content remains off-chain and is never directly committed to the blockchain. During verification, an authorized stakeholder recomputes the digest of the off-chain record and compares it with the corresponding on-chain anchor. This design supports tamper-evident verification while limiting the exposure of sensitive data. It is also consistent with privacy-preserving data-management principles because the blockchain stores evidence of integrity rather than clinical content itself.

The same off-chain Keccak-256 preparation layer is evaluated on three hardware targets. A workstation represents a high-performance consolidation node. An NVIDIA Jetson Nano represents an embedded edge gateway close to the data source. A Xilinx Zybo Zynq-7000 FPGA represents a low-power hardware/software co-design target where deterministic execution and energy efficiency are critical. Across these platforms, the study compares baseline software execution, memory-optimized software execution, CUDA-based execution, and FPGA-based acceleration using a custom Keccak-256 VHDL IP core.

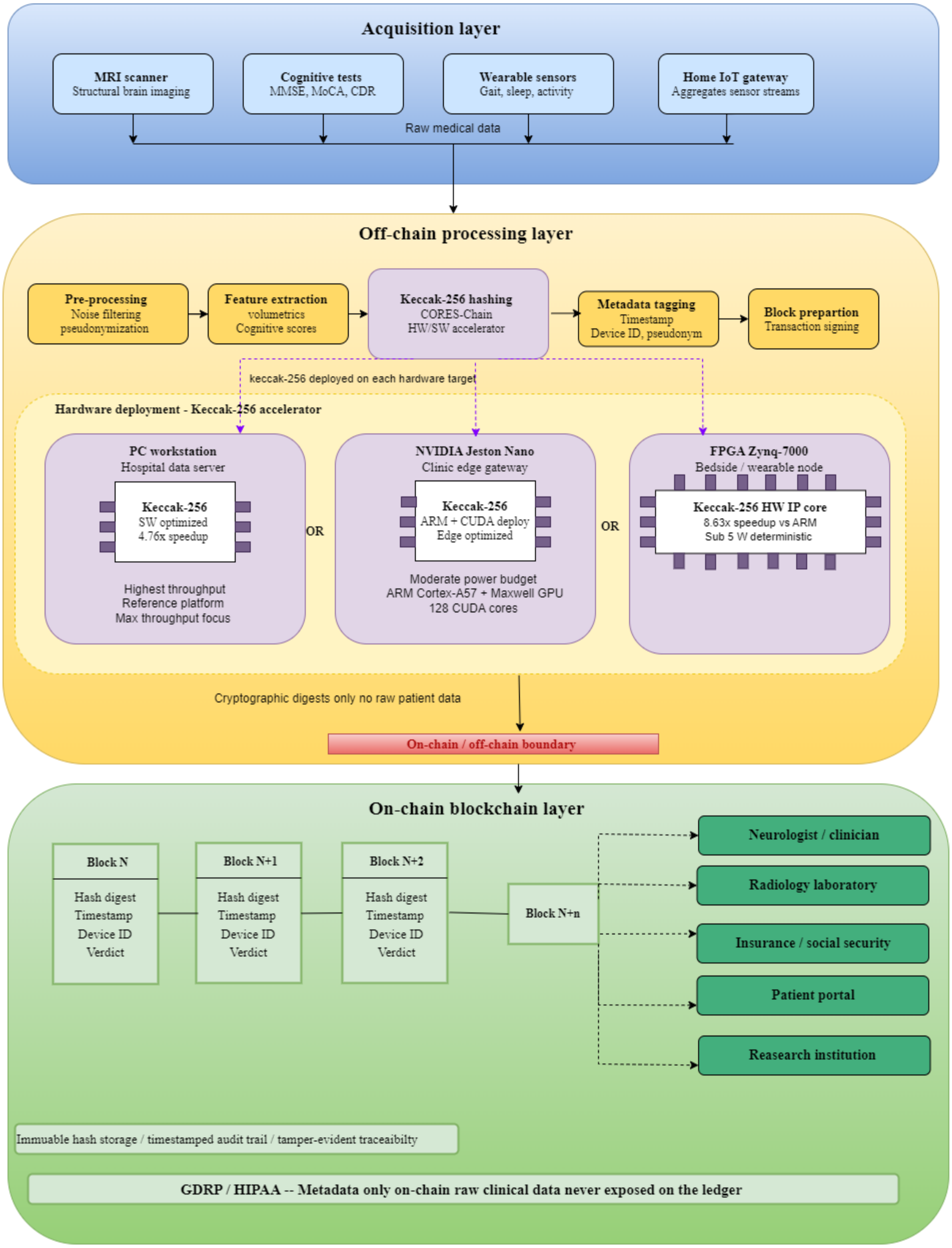

Figure 1: AURORA on-chain/off-chain anchoring architecture.

The remainder of this section details the threat model, the representative medical anchoring scenario, the off-chain preparation flow, the Keccak-256 hashing pipeline, the hardware/software implementation, and the platform-specific deployments.

3.2. Threat model and security assumptions

The proposed architecture assumes an adversary capable of observing, delaying, replaying, or attempting to tamper with records while they move between acquisition devices, edge gateways, and network-facing blockchain components. The main objective is therefore to reduce the exposure of sensitive records and intermediate processing stages by executing cryptographic preparation in a controlled off-chain environment before committing only finalized evidence to the blockchain layer.

AURORA does not assume that off-chain execution is universally trusted. Instead, the off-chain domain is considered a controlled environment whose security depends on the deployment configuration, including device protection, access control, secure storage, and communication safeguards. The blockchain layer provides an immutable anchoring mechanism for later verification, but it does not by itself guarantee the correctness of the off-chain preprocessing or the clinical validity of the records.

Under this model, the architecture provides integrity anchoring rather than full end-to-end clinical security. A record can later be verified by recomputing its Keccak-256 digest and comparing it with the digest committed to the blockchain layer. If the off-chain record is modified after anchoring, the recomputed digest will no longer match the on-chain value. Similarly, the timestamped sequence of anchors can help detect missing, reordered, or inconsistent records. However, the architecture does not prevent attacks occurring before the initial record is hashed, nor does it replace authentication, secure communication, access control, or regulatory compliance mechanisms.

This paper therefore focuses on the performance and implementation implications of the cryptographic preparation stage. Formal verification of the complete security protocol, network-level consensus analysis, and clinical deployment validation are outside the scope of the present work.

3.3. Representative use case: longitudinal medical data anchoring

To illustrate the motivation behind the architecture, we consider a longitudinal medical-data anchoring scenario inspired by Alzheimer's-related monitoring. This scenario is used as a representative use case only; the experimental evaluation in Section 4 does not use patient records, cognitive-test datasets, or medical imaging datasets.

In such a scenario, records may include structured cognitive scores, imaging-derived summaries, and wearable or home-sensor observations. These records are clinically meaningful because their value often depends on their ordering and evolution over time. A single isolated value may be less informative than a verified longitudinal sequence showing whether a trend, decline, or anomaly has occurred. Consequently, the integrity and chronological consistency of the record sequence are important.

AURORA supports this requirement by anchoring a digest of each off-chain record together with metadata such as acquisition time, device identifier, and record category. The clinical content remains under off-chain control, while the blockchain layer provides a tamper-evident audit trail. During verification, an authorized stakeholder can recompute the digest from the off-chain record and compare it with the anchored digest. This enables integrity checking without placing cognitive scores, imaging summaries, or sensor traces directly on-chain.

The three evaluated hardware platforms correspond to different deployment roles. The workstation can represent a hospital or research-site consolidation server. The Jetson Nano can represent a home or clinic edge gateway that aggregates records from nearby sensors or assessment devices. The Zynq-7000 FPGA can represent a low-power acquisition or bedside node where deterministic timing and power efficiency are more important than maximum aggregate throughput.

3.4. Off-chain preparation flow

The off-chain preparation flow is designed to provide a common workload across all evaluated platforms. A common genesis hash is first generated using Keccak-256. This value serves as the initial anchor for subsequent block-like records and ensures that all platforms execute comparable chained hashing operations.

For each input record, the preparation flow performs four steps. First, the record fields and metadata are serialized into a deterministic representation. Second, the serialized payload is concatenated with the previous digest or block reference, depending on the execution mode. Third, the resulting message is hashed using Keccak-256 to produce a 256-bit digest. Fourth, the digest is formatted and stored together with the metadata required for later verification.

The same logical flow is implemented on the workstation, the Jetson Nano, and the Zynq-7000 FPGA. This common structure enables direct comparison of execution time, throughput, execution-time variability, and power-normalized efficiency across heterogeneous hardware targets. The comparison is carried out entirely at the off-chain preparation layer; full network-level blockchain consensus and distributed deployment are left for future work.

3.5. Keccak-256 hashing pipeline

Figure 2 shows the Keccak-256 hashing pipeline used as the cryptographic core of AURORA. Keccak-256 transforms a variable-length input message into a fixed 256-bit digest through a sponge construction based on a 1600-bit internal state.

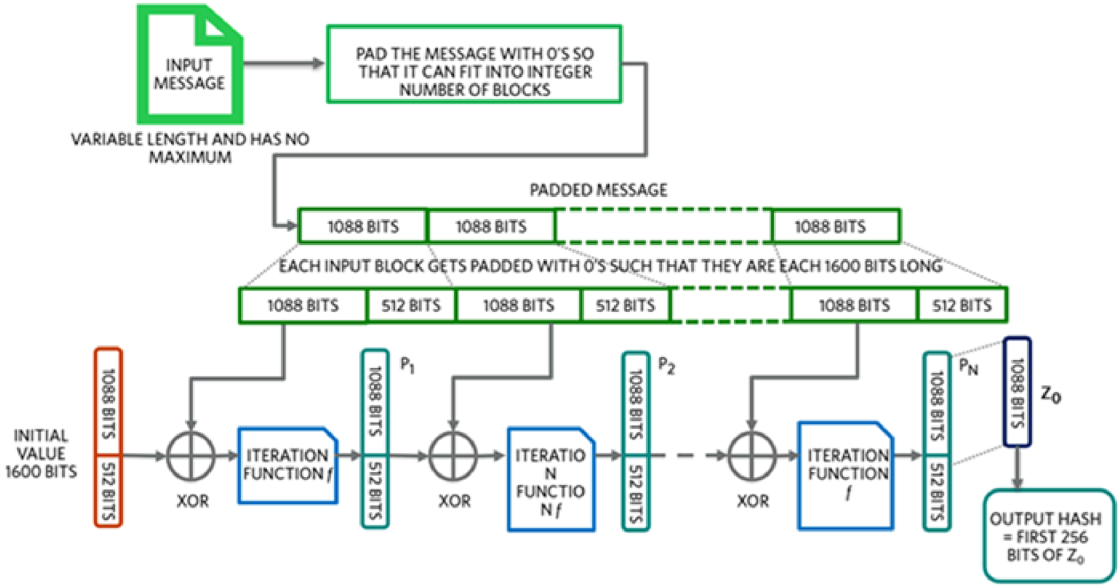

Figure 2: Keccak-256 hashing pipeline.

The input message is first padded and divided into blocks according to the rate parameter. For Keccak-256, the rate is $r = 1088$ bits and the capacity is $c = 512$ bits, giving a total internal state of 1600 bits. During the absorption phase, each message block is XORed with the rate portion of the internal state, after which the Keccak-$f[1600]$ permutation is applied. This permutation consists of the θ, ρ, π, χ, and ι transformations, which provide diffusion and non-linearity across the internal state. After all input blocks have been absorbed, the squeezing phase extracts the first 256 bits of the final state as the digest.

In this work, the term Keccak-256 refers to the Ethereum-compatible Keccak hashing primitive used for blockchain-oriented digest generation. The implementation is validated against reference outputs to ensure consistency across the C/C++, CUDA, and VHDL versions. Particular attention is given to byte ordering and digest formatting, since incorrect endianness can lead to mismatches between software and hardware implementations.

Figure 3 summarizes how the Keccak-256 digest-generation stage is embedded in the proposed off-chain preparation model.

```
genesis_block = {
    "timestamp": "YYYY-MM-DDTHH:MM:SSZ",
    "creator": "ENIS",
    "initialMessage": "Official launch of our blockchain!",
    "stateRoot": "0x0000000000000000000000000000000000000000",
    "bodyRoot": "0x0000000000000000000000000000000000000000",
    "nonce": 0,
    "version": "1.0"
}
```

Figure 3: Off-chain Keccak-256 preparation model.

3.6. Hardware/software implementation methodology

The implementation methodology follows a progressive optimization strategy. First, a reference software implementation is developed to establish functional correctness and baseline performance. Second, targeted software optimizations are introduced to reduce avoidable memory allocation, formatting overhead, and temporary object creation. Third, a CUDA implementation is developed to evaluate whether GPU parallelism is beneficial for the selected Keccak-256 workload. Fourth, a custom Keccak-256 hardware IP core is integrated into the programmable logic of the Zynq-7000 platform and compared with an ARM-only embedded baseline.

The profiling of the reference C++ implementation identified three dominant contributors to execution cost: the Keccak-f[1600] permutation, per-record serialization, and hexadecimal digest formatting. These observations guided the optimization strategy. On the software side, the implementation uses pre-allocated buffers, lookup-table-based hexadecimal encoding, lightweight timestamp generation, and reduced intermediate allocations. On the hardware side, the Keccak permutation is mapped to a dedicated VHDL IP core controlled by the ARM processor through an AXI4-Lite interface.

The same input size, hashing workload, and repetition protocol are used across platforms. This ensures that differences in the results reflect implementation and hardware behavior rather than differences in workload definition.

3.7. Platform-specific implementation

3.7.1. Workstation implementation

The workstation implementation is developed in C++ and executed in a single-threaded configuration for the CPU baseline. The baseline version uses a standard Keccak-256 software routine, conventional string operations for block-field assembly, and iterative formatting for digest conversion. The memory-optimized version introduces four modifications: pre-allocation of block-field buffers, lookup-table-based hexadecimal encoding, lightweight timestamp generation, and removal of redundant intermediate string temporaries.

A CUDA version is also implemented on the workstation GPU. Its purpose is not to claim optimal GPU hashing performance, but to evaluate whether coarse-grained GPU parallelism provides an advantage for the selected Keccak-256 preparation workload when compared with optimized CPU execution.

3.7.2. Jetson Nano implementation

The Jetson Nano implementation uses the same logical pipeline as the workstation. The ARM Cortex-A57 processor executes both the baseline and memory-optimized software versions, while the embedded Maxwell GPU executes the CUDA version. This platform is included because it represents a realistic low-power edge gateway and enables comparison between embedded GPU execution and CPU-based execution under constrained hardware resources.

3.7.3. *Zybo Zynq-7000 FPGA implementation*

Two Zynq-7000 variants are evaluated. The first is an ARM-only implementation, where the Keccak-256 preparation pipeline runs entirely on the ARM Cortex-A9 processor. This configuration provides the embedded software baseline.

The second variant integrates a custom Keccak-256 IP core in programmable logic. The ARM processor handles orchestration, input preparation, control signaling, and output collection, while the programmable logic executes the Keccak-f[1600] permutation. The accelerator is interfaced through AXI4-Lite, which simplifies integration with the processing system while allowing direct comparison with the ARM-only baseline.

During hardware validation, particular attention is given to digest correctness. Byte ordering and nibble orientation are checked against software reference outputs, since endianness mismatches are common when mapping cryptographic algorithms from software to hardware. The final corrected implementation produces outputs consistent with the C/C++ reference implementation.

Figure 4 illustrates the cross-platform execution flow used to compare workstation, Jetson Nano, and Zynq-7000 FPGA implementations.

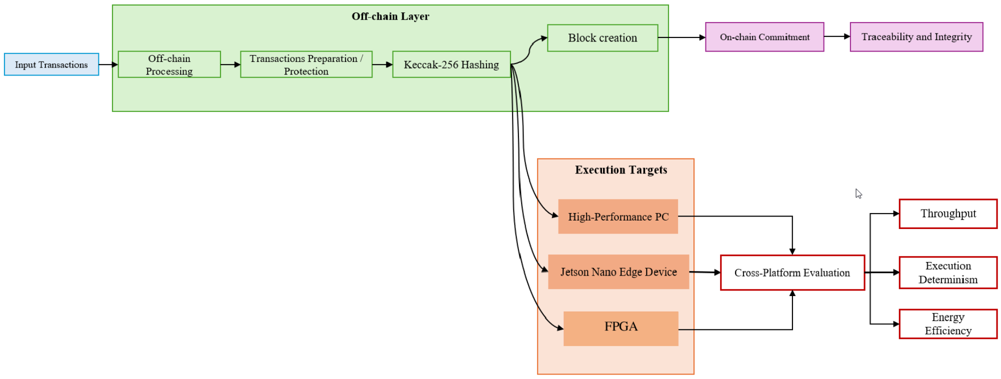

Figure 4: Cross-platform AURORA execution flow.

3.8. Prototype PoW-oriented FPGA extension

In addition to the main Keccak-256 preparation pipeline, a prototype Proof-of-Work-oriented FPGA extension was developed to explore how nonce iteration and difficulty checking could be integrated with the hardware hashing core. This extension is not used as the basis for the quantitative throughput comparison reported in Section 4; the main experimental results correspond to full Keccak-256 hashing over 10^6 inputs.

The prototype contains three hardware components. The `pow_sequencer_core` prepares input blocks, inserts the nonce and padding, controls the handshake with the Keccak core, and manages the nonce-increment loop. The `keccak_core` absorbs input blocks, applies the Keccak-f[1600] permutation, and produces the 256-bit digest. The `pow_difficulty_checker` evaluates whether the resulting hash satisfies the configured difficulty condition. Figure 5 shows the finite-state machine used in the prototype PoW-oriented FPGA extension.

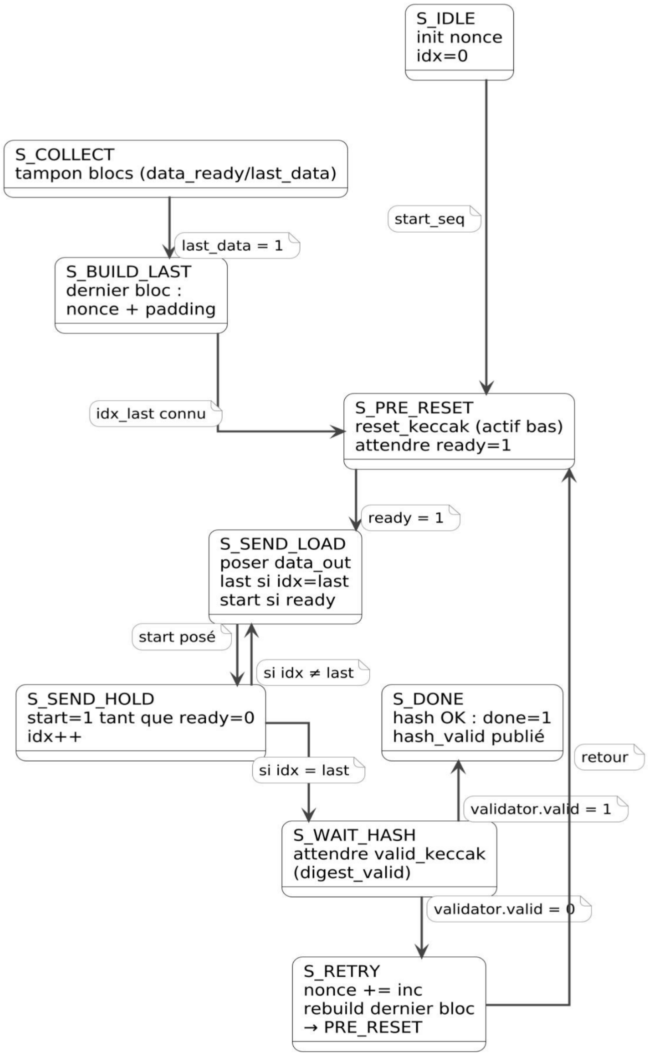

Figure 5: PoW-oriented FPGA sequencer.

The sequencer finite-state machine includes states for input collection, last-block construction, Keccak reset, block loading, hash waiting, nonce retry, and completion. This prototype confirms the feasibility of coupling nonce management and difficulty checking with the Keccak hardware core. However, because the present paper focuses on the stable and reproducible Keccak-256 preparation workload, full PoW nonce-search benchmarking is treated as future work rather than as a main quantitative contribution.

Figure 6 presents the implemented Keccak-256 hardware design on the Zynq-7000 FPGA platform.

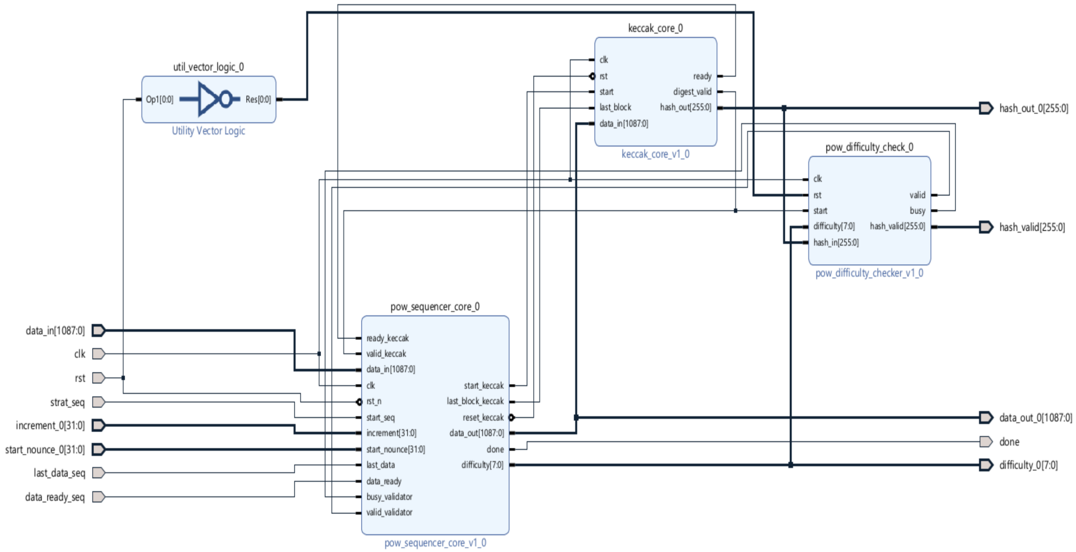

Figure 6: Zynq-7000 Keccak-256 hardware design.

4. Experimental Results

This section evaluates the off-chain Keccak-256 preparation layer of the proposed AURORA architecture. The experiments are conducted on three hardware targets: a high-performance workstation, an NVIDIA Jetson Nano edge platform, and a Xilinx Zybo Zynq-7000 FPGA. The objective is to quantify execution time, throughput, execution-time variability, and power-normalized efficiency for a common Keccak-256 workload across heterogeneous platforms.

All configurations are evaluated using a workload of 10^6 Keccak-256 hashes and ten independent runs per configuration. For each implementation, we report the mean execution time, standard deviation, coefficient of variation (CV), throughput, and estimated power-normalized efficiency. The coefficient of variation is used to characterize execution-time stability across repeated runs. Throughput is expressed in hashes per second (H/s), while power-normalized efficiency is expressed in hashes per second per watt (H/s/W).

It is important to note that the quantitative results reported in this section correspond to the stable Keccak-256 preparation workload. They do not represent a full network-level blockchain deployment, nor a complete distributed consensus benchmark. The Proof-of-Work-oriented FPGA extension described in Section 3.8 is treated as a prototype extension and is not used as the basis for the main quantitative comparison.

4.1. Experimental setup and power-normalization model

Table 2 summarizes the main characteristics of the evaluated hardware platforms. The workstation provides a high-throughput reference platform for optimized CPU execution and desktop GPU evaluation. The Jetson Nano represents a low-power embedded edge gateway. The Zybo Zynq-7000 platform combines an ARM Cortex-A9 processing system with programmable logic, enabling a direct comparison between ARM-only execution and FPGA-based hardware/software co-design.

Table 2
Hardware platforms used in the experimental evaluation

Platform	CPU Characteristics	GPU/FPGA Resource	Power Envelope
Dell Inspiron 16 7630	16-core CPU, up to 5 GHz	RTX 4060 8 GB	45 W + 115 W
Jetson Nano 4 GB	Quad-core ARM Cortex-A57, 1.4 GHz	128-core Maxwell GPU	Max. 20 W
Zybo Zynq-7000	Dual-core ARM Cortex-A9, ∼667 MHz	Artix-7 PL	<5 W

CPU: Central Processing Unit; *GPU*: Graphics Processing Unit; *FPGA*: Field-Programmable Gate Array; *ARM*: Advanced RISC Machine processor architecture; *RTX*: NVIDIA Ray Tracing GPU series; *PL*: Programmable Logic (Zynq FPGA fabric); *W*: Watt (unit of power consumption).

Because inline electrical measurements were not available for all configurations, the H/s/W values are reported as estimated power-normalized indicators rather than direct electrical measurements. The purpose of this normalization is to support relative comparison under consistent assumptions, not to claim absolute energy consumption.

The normalization envelopes used in the comparative table are: 10 W for CPU-only workstation execution, 115 W for RTX 4060 GPU execution, 10 W for Jetson Nano CPU execution, 15 W for Jetson Nano GPU execution, and 5 W for the FPGA-based configurations. These values are used only for comparative normalization. Future work will complement this analysis with inline power sensing to obtain measured energy-per-hash values.

4.2. Workstation results

Table 3 reports the workstation results. The baseline C++ implementation requires 13.673 s on average to compute 10^6 hashes, corresponding to 73,279 H/s. The memory-optimized implementation reduces the mean execution time to 2.869 s, increasing throughput to 349,303 H/s. This corresponds to a 4.76× speedup over the baseline.

The improvement is mainly due to the reduction of avoidable software overheads: pre-allocated buffers reduce repeated memory allocation, lookup-table-based hexadecimal encoding avoids iterative `sprintf`-style formatting, lightweight timestamp generation reduces formatting cost, and redundant intermediate string temporaries are removed. The CV remains close to the baseline, indicating that the optimization improves throughput without increasing timing variability.

The CUDA implementation on the RTX 4060 is slower than both CPU variants, with a mean time of 72.06 s. This result should not be interpreted as a universal limitation of GPU hashing, but rather as an indication that the selected Keccak-256 preparation workload and implementation do not benefit from coarse-grained GPU execution. The likely causes include kernel-launch overhead, host-device transfer cost, register pressure caused by the 1600-bit Keccak state, and memory-access overhead during digest handling.

Table 3
Workstation Keccak-256 results (10^6 hashes, $n = 10$)

Implementation	Mean (s)	Std. Dev. (s)	CV (%)	H/s
Baseline C++	13.673	0.647	4.73	73,279
Memory-optimized C++	2.869	0.146	5.08	349,303
CUDA RTX 4060	72.060	3.800	5.27	13,878
Speedup, optimized vs. baseline: 4.76×				

H/s: Hashes per second (throughput metric); *Mean*: Average execution time over $n = 10$ runs; *Std. Dev.*: Standard deviation of execution time; *CV*: Coefficient of Variation, ratio of standard deviation to mean (stability indicator); *CUDA*: Compute Unified Device Architecture (NVIDIA parallel computing platform); *RTX 4060*: NVIDIA GeForce RTX 4060 discrete GPU; *C++*: ISO-standard compiled programming language used for software implementations; *Speedup*: Ratio of baseline execution time to optimized execution time.

4.3. Jetson Nano results

Table 4 reports the results obtained on the Jetson Nano. The baseline ARM implementation requires 23.650 s on average, corresponding to 42,283 H/s. The memory-optimized ARM version reduces the mean execution time to 11.823 s and increases throughput to 84,603 H/s, giving a 2.00× speedup over the baseline.

The lower absolute throughput compared with the workstation is expected because the Jetson Nano uses an embedded ARM Cortex-A57 processor with lower clock frequency and lower memory bandwidth than the workstation CPU. However, the CV values for the ARM implementations are lower than those observed on the workstation, suggesting more stable execution under the tested embedded environment.

The CUDA implementation on the Jetson Nano performs poorly for this workload, with a mean time of 407.385 s and a CV of 11.17%. This confirms that, under the tested implementation, the embedded Maxwell GPU is not suitable for this specific Keccak-256 preparation workload. The overhead of launching kernels and managing GPU execution dominates the benefit of parallelism.

Table 4
Jetson Nano Keccak-256 results (10^6 hashes, $n = 10$)

Implementation	Mean (s)	Std. Dev. (s)	CV (%)	H/s
Baseline ARM	23.650	0.384	1.62	42,283
Memory-optimized ARM	11.823	0.072	0.61	84,603
CUDA Maxwell GPU	407.385	45.496	11.17	2,455
Speedup, optimized vs. baseline: 2.00×				

H/s: Hashes per second (throughput metric); *Mean*: Average execution time over $n = 10$ runs; *Std. Dev.*: Standard deviation of execution time; *CV*: Coefficient of Variation (stability indicator); *ARM*: Advanced RISC Machine processor architecture (Cortex-A57 on Jetson Nano); *CUDA*: Compute Unified Device Architecture (NVIDIA parallel computing platform); *Maxwell*: NVIDIA Maxwell GPU microarchitecture embedded in the Jetson Nano; *Speedup*: Ratio of baseline execution time to optimized execution time.

4.4. Zybo Zynq-7000 FPGA results

Table 5 summarizes the Zynq-7000 results. The ARM-only implementation requires 151.143 s on average to compute 10^6 hashes, corresponding to 6,617 H/s. This low throughput reflects the limited clock frequency and embedded nature of the ARM Cortex-A9 processing system.

When the custom Keccak-256 IP core is instantiated in the programmable logic and controlled by the ARM processor, the mean execution time decreases to 16.475 s. This corresponds to a throughput of 60,704 H/s and an 9.17× speedup over the ARM-only baseline. The CV is reduced to 0.013%, showing highly stable timing behavior. **[VERIFY: the previously reported 8.63× speedup is inconsistent with the reported means (151.143/16.475 = 9.17); confirm against raw logs before submission.]** This low variability is a key advantage of FPGA-based execution for embedded systems where deterministic response is more important than maximum aggregate throughput.

Although the FPGA configuration does not reach the absolute throughput of the optimized workstation CPU, it provides a favorable trade-off for low-power embedded deployment. In particular, it combines hardware-level acceleration, deterministic execution, and a low normalization envelope compared with GPU-based configurations.

Table 5
Zybo Zynq-7000 FPGA Keccak-256 results (10^6 hashes, $n = 10$)

Implementation	Mean (s)	Std. Dev. (s)	CV (%)	H/s
ARM software only	151.143	0.029	0.019	6,617
ARM + Keccak-256 IP core	16.475	0.021	0.013	60,704
Speedup, IP core vs. ARM only: 9.17×				

H/s: Hashes per second (throughput metric); *Mean*: Average execution time over $n = 10$ runs; *Std. Dev.*: Standard deviation of execution time; *CV*: Coefficient of Variation (stability indicator); *ARM*: Advanced RISC Machine processor (Cortex-A9 on Zybo Zynq-7000); *FPGA*: Field-Programmable Gate Array (Artix-7 programmable logic fabric); *IP core*: Intellectual Property core, a reusable hardware block implementing Keccak-256 in the FPGA fabric; *Zynq-7000*: Xilinx SoC combining ARM processor and Artix-7 FPGA fabric; *Speedup*: Ratio of ARM-only execution time to IP-core-accelerated execution time.

4.5. Anchoring-oriented latency interpretation

For blockchain anchoring modes that do not require nonce search, such as Proof-of-Authority or permissioned validation settings, the dominant local cryptographic cost is the computation of the record digest and its preparation for anchoring. The measurements above can therefore be interpreted as local digest-generation latency estimates, excluding network propagation, validator scheduling, smart-contract execution, and block-finalization delay.

With the optimized workstation implementation, one Keccak-256 digest is produced in approximately 2.9 μs on average. On the Jetson Nano, the optimized ARM implementation requires approximately 11.8 μs per digest. On the Zynq-7000 FPGA with the Keccak-256 IP core, the average digest time is approximately 16.5 μs, with very low variability. This suggests that the optimized workstation is best suited for high-throughput consolidation nodes, while the FPGA configuration is better suited for deterministic low-power acquisition or gateway nodes.

These values should not be interpreted as full PoA or full blockchain transaction latency. They represent only the local cryptographic preparation component of an anchoring pipeline. End-to-end blockchain latency requires additional measurement of transaction creation, communication, validation, storage, and confirmation time.

4.6. Cross-platform comparative analysis

Table 6 consolidates the cross-platform results. The memory-optimized workstation implementation achieves the highest absolute throughput, reaching 349,303 H/s. Under the adopted normalization envelope, it also obtains the highest estimated H/s/W value. However, this result depends on the selected CPU normalization assumption and should be interpreted as a comparative indicator rather than as a direct electrical measurement.

The FPGA-based HW/SW co-design achieves 60,704 H/s, which is lower than the optimized workstation but substantially higher than the ARM-only FPGA baseline. Its main advantage is not maximum throughput, but deterministic execution and favorable low-power embedded behavior. The CV of 0.013% is the lowest among all configurations, making the FPGA target attractive for time-constrained embedded anchoring tasks.

The Jetson Nano optimized ARM implementation achieves an intermediate operating point, with 84,603 H/s and low timing variability. This makes it suitable for edge-gateway scenarios where moderate throughput and software flexibility are required. In contrast, the GPU implementations on both the workstation and Jetson Nano are inefficient for the tested workload. They show lower throughput and lower estimated power-normalized efficiency than their CPU counterparts.

Table 6
Cross-platform Keccak-256 performance summary (10^6 hashes)

Platform	Implementation	Time (s)	H/s	P_{norm} (W)	Eff. (H/s/W)
Workstation	Baseline C++	13.67	73,279	10	7,328
Workstation	Mem.-opt. C++	2.87	349,303	10	**34,930**
Workstation	CUDA RTX 4060	72.06	13,878	115	121
Jetson Nano	Baseline ARM	23.65	42,283	10	4,228
Jetson Nano	Mem.-opt. ARM	11.82	84,603	10	8,460
Jetson Nano	CUDA Maxwell	407.38	2,455	15	164
Zybo Zynq-7000	ARM only	151.14	6,617	5	1,323
Zybo Zynq-7000	**ARM + IP core**	**16.47**	**60,704**	**5**	**12,141**

H/s: Hashes per second (throughput metric); *Eff.*: Energy efficiency expressed as hashes per second per watt (H/s/W); P_{norm}: Normalized power envelope used for efficiency comparison; not a direct electrical measurement; *ARM*: Advanced RISC Machine processor architecture; *CUDA*: Compute Unified Device Architecture (NVIDIA parallel computing platform); *RTX 4060*: NVIDIA GeForce RTX 4060 discrete GPU (workstation); *Maxwell*: NVIDIA Maxwell GPU microarchitecture embedded in the Jetson Nano; *IP core*: Reusable Keccak-256 hardware block implemented in the Zynq-7000 FPGA fabric; *Mem.-opt.*: Memory-optimized software implementation.

4.7. Engineering implications

The results highlight three engineering implications for embedded blockchain-based medical data anchoring.

First, software optimization is highly effective when the target platform has sufficient CPU resources. On the workstation, memory-aware implementation reduces execution time by 4.76× compared with the baseline. On the Jetson Nano, the same optimization strategy provides a 2.00× improvement, confirming that buffer management and formatting overhead are important contributors to Keccak-256 preparation cost.

Second, FPGA-based HW/SW co-design provides the best timing determinism among the evaluated embedded targets. The custom Keccak-256 IP core reduces execution time by 9.17× compared with ARM-only execution on the Zynq-7000 platform and achieves the lowest CV. This makes the FPGA configuration suitable for low-power nodes where predictable latency is required.

Third, GPU execution is not advantageous for the tested Keccak-256 preparation workload. Both the RTX 4060 and Jetson Nano Maxwell GPU variants underperform their corresponding CPU implementations. This result suggests that, for this workload size and implementation structure, the overhead of GPU execution outweighs the benefit of parallelism.

Overall, the optimized workstation is the best choice for high-throughput consolidation, the Jetson Nano offers a flexible edge-gateway compromise, and the FPGA-based HW/SW co-design provides the most deterministic embedded execution profile. These findings support the use of heterogeneous deployment strategies in which the same off-chain blockchain preparation layer is mapped to different hardware targets according to throughput, latency, and power constraints.

5. Discussion

5.1. Cross-platform trade-offs, architectural insights, and deployment guidelines

The cross-platform evaluation reported in Section 4 provides several insights into the deployment of Keccak-256-based off-chain preparation pipelines on heterogeneous hardware. The first observation concerns the limited benefit of GPU execution for the evaluated workload. On both the desktop RTX 4060 and the embedded Jetson Nano Maxwell GPU, the CUDA implementations are slower than their corresponding CPU implementations despite the higher nominal parallel-processing capability of the GPUs. This result suggests that the tested Keccak-256 preparation workload does not map efficiently to coarse-grained GPU execution.

Several implementation-level factors may explain this behavior. First, kernel-launch overhead becomes significant when the amount of useful computation per hash is relatively small. Second, the 1600-bit Keccak internal state, represented as 25 64-bit words, increases register pressure and may lead to memory spills. Third, host-device transfers and digest-handling overheads further reduce the benefit of GPU parallelism. These results are consistent with prior observations that FPGA-based acceleration can be more suitable than GPU execution for certain blockchain-related cryptographic workloads (Ktari et al., 2024b). Nevertheless, this conclusion should be interpreted within the scope of the tested implementation. More specialized GPU strategies, such as persistent kernels, warp-level primitives, batching optimizations, or shared-memory-aware implementations, could improve performance and should be explored in future work.

The second observation is the importance of memory-aware software optimization. On the workstation, the optimized C++ implementation achieves a 4.76× speedup over the baseline, while on the Jetson Nano the optimized ARM implementation achieves a 2.00× speedup. These gains are obtained without changing the underlying hardware and are mainly due to the reduction of non-cryptographic overheads, including repeated memory allocation, block-field serialization, timestamp formatting, and hexadecimal digest conversion. This confirms that, in practical blockchain-preparation pipelines, the cost is not limited to the Keccak-$f[1600]$ permutation itself. Formatting and data-management operations can become significant contributors to runtime if they are not carefully implemented.

The FPGA results provide the strongest evidence for the relevance of hardware/software co-design in embedded blockchain anchoring. On the Zynq-7000 platform, the custom Keccak-256 IP core reduces execution time by 9.17× compared with the ARM-only baseline. More importantly, the FPGA configuration exhibits the lowest execution-time variability among all evaluated platforms, with a coefficient of variation of 0.013%. This deterministic behavior is particularly relevant for embedded and edge applications where predictable latency is more important than peak throughput.

Under the adopted power-normalization model, the FPGA-based HW/SW co-design achieves 12,141 H/s/W, compared with 1,323 H/s/W for the ARM-only FPGA baseline and 8,460 H/s/W for the optimized Jetson Nano ARM implementation. The optimized workstation CPU remains the highest-throughput configuration and also obtains the highest estimated H/s/W value under the selected normalization assumptions. However, this operating point corresponds to a workstation-class environment and is not directly representative of low-power embedded deployment. The FPGA configuration therefore offers a more suitable profile for nodes where deterministic timing, hardware-level acceleration, and constrained power operation are primary requirements.

Figure 7 summarizes the cross-platform performance and power-normalized trade-offs.

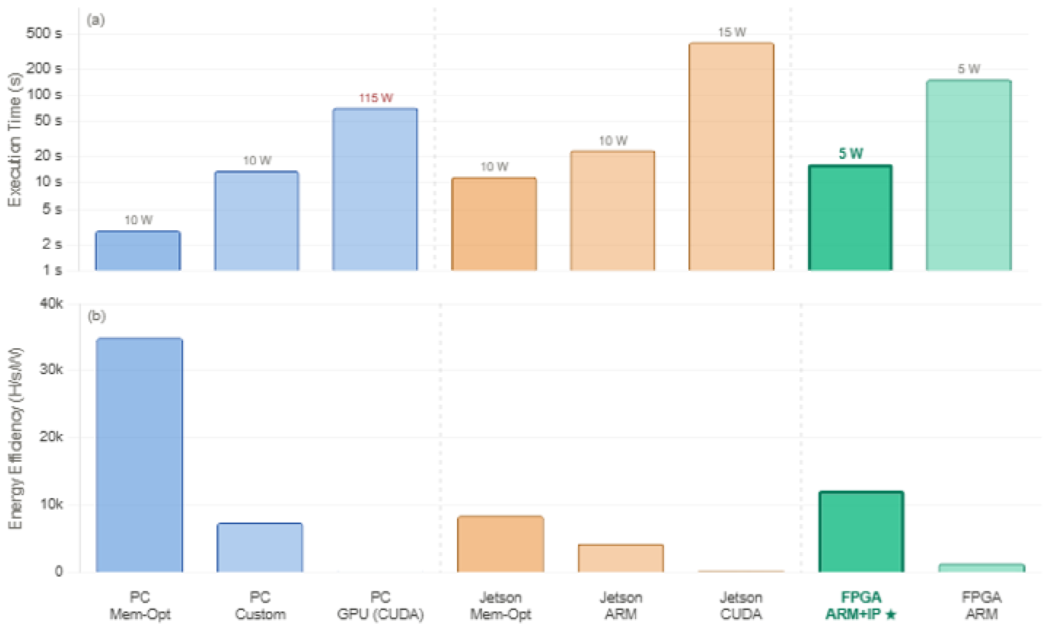

Figure 7: Performance and efficiency trade-offs.

Relative to prior FPGA-based blockchain acceleration studies, the contribution of AURORA lies in its cross-platform and implementation-oriented evaluation. Ktari et al. (2024b) and Frikha et al. (2021) demonstrate the value of FPGA acceleration for blockchain-related processing and consensus-oriented components. In contrast, the present work focuses on the off-chain cryptographic preparation layer required before blockchain anchoring and evaluates the same Keccak-256 workload across optimized CPU software, embedded ARM execution, CUDA implementations, and FPGA-based HW/SW co-design. This enables a more direct comparison of deployment options for heterogeneous edge environments.

The results also operationalize the motivation behind hybrid on-chain/off-chain architectures discussed by Eberhardt and Tai (2017) and Wöhrer and Zdun (2021). Rather than storing sensitive or large records directly on-chain, AURORA keeps the records off-chain and prepares cryptographic digests for anchoring. The experimental findings show that this preparation layer can be mapped to different hardware targets depending on deployment constraints: workstation-class systems for high-throughput consolidation, Jetson Nano-class devices for flexible edge gateways, and FPGA-based co-design for deterministic low-power nodes.

5.2. Implications for longitudinal medical data anchoring

The representative medical scenario introduced in Section 3.3 helps interpret how the evaluated platforms could be used in a longitudinal data-anchoring chain. In such a chain, different hardware nodes may have different roles. The memory-optimized workstation configuration, which provides the highest absolute throughput, is well suited to a hospital or research-site consolidation server that processes records from multiple patients. In this setting, mains power is available and aggregate throughput is the dominant requirement.

The Jetson Nano represents an intermediate deployment point. Its optimized ARM implementation provides moderate throughput with relatively low execution-time variability, making it suitable for a home or clinic edge gateway. Such a gateway could aggregate records from nearby devices, prepare digests, and forward anchoring metadata to a blockchain-facing service.

The FPGA-based HW/SW co-design is most relevant for low-power acquisition or bedside nodes. Although its throughput is lower than the optimized workstation CPU, its deterministic timing and low normalization envelope make it attractive for settings where predictable response and constrained power are more important than aggregate throughput. This profile is relevant for devices that periodically hash small records, sensor windows, or summarized measurements before transmission to a gateway.

This mapping should be understood as an applicability analysis rather than as clinical validation. A real medical deployment would require additional steps that are outside the scope of this paper. These include secure storage of off-chain records, patient authentication, access-control policies, encryption before and after hashing, metadata minimization, and compliance analysis under frameworks such as GDPR or HIPAA. In addition, the anchoring metadata itself must be carefully designed so that timestamps, device identifiers, or record frequencies do not reveal sensitive information or allow indirect re-identification.

The workload used in this study, namely repeated Keccak-256 hashing over 10^6 inputs, provides a controlled basis for comparing hardware platforms. However, real medical workloads may differ substantially. Cognitive-test records are typically small and structured, while imaging data or continuous sensor streams may require chunked, hierarchical, or streaming hashing strategies. The relative ranking observed in this paper provides useful engineering guidance, but it should be validated on realistic medical record sizes and acquisition frequencies before deployment.

5.3. Threats to validity

Several threats to validity should be considered when interpreting the results.

First, with respect to scope validity, the quantitative evaluation focuses on the Keccak-256 off-chain preparation layer rather than on a full network-level blockchain deployment. Transaction propagation, validator scheduling, smart-contract execution, block finalization, and distributed consensus latency are not measured in the present study. The reported results therefore characterize the local cryptographic preparation cost, not the end-to-end blockchain confirmation time.

Second, the Proof-of-Work-oriented FPGA extension is treated as a prototype and is not included in the main quantitative comparison. The stable experimental results correspond to repeated Keccak-256 hashing and preparation over 10^6 inputs. Full nonce-search benchmarking and complete PoW-loop validation are left for future work.

Third, with respect to measurement validity, power-normalized efficiency values are derived from representative normalization envelopes rather than inline electrical measurements. These values support relative comparison under consistent assumptions, but they should not be interpreted as precise physical energy measurements. Future work should include direct power monitoring, for example using external power meters or sensor-based instrumentation, in order to report measured energy per hash and validate the H/s/W ranking under real electrical conditions.

Fourth, with respect to external validity, the experiments focus on Keccak-256 in a blockchain-oriented processing context. Other cryptographic primitives, such as SHA-256, Blake2, Poseidon, or domain-specific authenticated data structures, may exhibit different behavior across CPU, GPU, and FPGA platforms. The conclusions should therefore be generalized cautiously beyond Keccak-256-based anchoring workloads.

Fifth, with respect to implementation validity, the GPU results reflect the CUDA implementations evaluated in this study. They should not be interpreted as a definitive upper bound on GPU-based Keccak-256 performance. More advanced GPU implementations may reduce overhead through batching, persistent kernels, or architecture-specific optimization.

Finally, with respect to the medical anchoring scenario, no patient cohort, cognitive-assessment corpus, imaging dataset, or regulatory review was included in the experimental evaluation. The Alzheimer's-related scenario is used to motivate longitudinal medical data anchoring, not to claim diagnostic, clinical, or regulatory readiness. Before any clinical pilot, the architecture would need to be evaluated using realistic medical workloads and integrated with secure data-management, consent, access-control, and compliance mechanisms.

Despite these limitations, the reported results provide useful engineering evidence for selecting hardware targets in embedded blockchain-based data anchoring. They show that software optimization is essential on CPU platforms, that FPGA-based HW/SW co-design provides strong deterministic behavior for constrained nodes, and that GPU execution is not automatically advantageous for Keccak-256 preparation workloads.

6. Conclusion

This paper presented AURORA, an energy-aware embedded on-chain/off-chain architecture for blockchain-based medical data anchoring. The proposed approach separates the off-chain cryptographic preparation layer from the network-facing anchoring layer, allowing sensitive records to remain off-chain while Keccak-256 digests and associated metadata are prepared for later blockchain verification. The study focused on the local execution cost of this preparation layer and evaluated it across heterogeneous hardware targets, including a workstation, an NVIDIA Jetson Nano edge platform, and a Xilinx Zybo Zynq-7000 FPGA.

The experimental results show that memory-aware software optimization significantly improves CPU execution. On the workstation, the optimized C++ implementation achieves a 4.76× speedup over the baseline, while on the Jetson Nano the optimized ARM implementation achieves a 2.00× speedup. The FPGA-based hardware/software co-design provides an 9.17× speedup over the ARM-only embedded baseline and exhibits the lowest execution-time variability among all evaluated configurations. These results indicate that optimized workstation execution is best suited for high-throughput consolidation nodes, whereas FPGA-based acceleration provides a more deterministic profile for low-power embedded nodes. In contrast, the evaluated GPU implementations are not competitive for the tested Keccak-256 preparation workload, mainly because GPU execution overhead outweighs the benefit of parallelism under the selected workload and implementation conditions.

The main contribution of this work lies in combining an embedded on-chain/off-chain anchoring architecture, a practical Keccak-256 hardware/software acceleration path, and a cross-platform evaluation methodology based on execution time, throughput, execution-time variability, and estimated power-normalized efficiency. The results provide engineering guidance for selecting appropriate hardware targets in embedded blockchain-based data anchoring systems. Workstation-class platforms are appropriate when aggregate throughput is the main requirement, Jetson Nano-class platforms provide a flexible edge-gateway compromise, and FPGA-based co-design is particularly attractive when deterministic timing and constrained power operation are required.

The architecture was illustrated through a longitudinal medical-data anchoring scenario inspired by Alzheimer's-related monitoring. In this scenario, cognitive-assessment records, imaging-derived summaries, or sensor-based observations remain off-chain, while their cryptographic digests can be anchored for later integrity verification. This scenario should be interpreted as an applicability case rather than as clinical validation. The present work does not claim diagnostic capability, regulatory readiness, or deployment in a real clinical environment.

Several extensions remain open. Future work will integrate the preparation layer with a complete blockchain deployment in order to measure transaction creation, communication, validation, and final anchoring latency. Direct inline electrical power measurements will also be required to replace the current normalization-based H/s/W estimates with measured energy-per-hash values. Additional work will investigate optimized GPU implementations, alternative cryptographic primitives, larger and more realistic record sizes, and multi-node deployment settings. The prototype PoW-oriented FPGA extension will be further stabilized and benchmarked separately. Finally, moving the medical anchoring scenario toward a practical pilot would require realistic cognitive, imaging, and sensor workloads, secure off-chain storage, access-control mechanisms, metadata minimization, and system-level compliance analysis under frameworks such as GDPR and HIPAA.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics approval and consent to participate

Not applicable. This study did not involve human participants, human tissue, identifiable personal data, clinical intervention, or patient recruitment. The medical-data anchoring scenario discussed in the manuscript is used only as a representative applicability case.

Consent for publication

Not applicable. This manuscript does not contain identifiable personal data, patient images, or individual clinical information.

Data availability

The benchmark data supporting the findings of this study, including execution-time measurements, throughput values, coefficient-of-variation calculations, and derived power-normalized efficiency values, are available from the corresponding author upon reasonable request.

Code availability

The source code and hardware-description materials used in this study, including the C++ benchmark implementation, CUDA implementation, and VHDL Keccak-256 IP validation files, are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Hanedi Loukil: Conceptualization, Methodology, Writing – Original Draft. **Amira Talha:** Methodology, Writing – Review & Editing. **[VERIFY: author-block credit and CRediT statement disagreed; confirm Talha's roles.]** **Tarek Frikha:** Methodology, Validation, Writing – Review & Editing. **Mathurin Conan:** Hardware Implementation, Writing – Review & Editing. **Clément Loiseau:** Hardware Implementation, Writing – Review & Editing. **Samir Bouaziz:** Supervision, Writing – Review & Editing.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used AI-assisted tools to support English language editing, readability improvement, and manuscript polishing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

Abrar, I., Sheikh, J., 2024. Current trends of blockchain technology: architecture, applications, challenges, and opportunities. Discov. Internet Things 4, 7. doi:10.1007/s43926-024-00058-5.

Ahmadjee, S., Mera-Gómez, C., Bahsoon, R., Kazman, R., 2022. A study on blockchain architecture design decisions and their security attacks and threats. ACM Trans. Softw. Eng. Methodol. 31. doi:10.1145/3502740.

Ahmed, S., et al., 2024. Blockchain-based smart contract model for securing healthcare transactions using consumer electronics and mobile-edge computing. IEEE Trans. Consum. Electron. doi:10.1109/TCE.2024.3358975.

Allouche, M., Frikha, T., Mitrea, M., Memmi, G., Chaabane, F., 2021. Lightweight blockchain processing. Case study: scanned document tracking on Tezos blockchain. Applied Sciences 11, 7169. doi:10.3390/app11157169.

AMD (Xilinx), 2025. Zynq-7000 SoCs product page. https://www.amd.com/en/products/adaptive-socs-and-fpgas/soc/zynq-7000.html. Accessed: August 2025.

Chaabane, F., Ktari, J., Frikha, T., Hamam, H., 2022. Low power blockchained e-vote platform for university environment. Future Internet 14, 269. doi:10.3390/fi14090269.

De Aguiar, E., Faiçal, B., Krishnamachari, B., Ueyama, J., 2021. A survey of blockchain-based strategies for healthcare. ACM Comput. Surv. 53, 1–27. doi:10.1145/3376915.

Dorri, A., Kanhere, S., Jurdak, R., 2019. Hybrid blockchain architectures for the Internet of Things: energy and scalability considerations. IEEE Internet Things J. 6, 4946–4960. doi:10.1109/JIOT.2019.2900241.

Eberhardt, J., Tai, S., 2017. On or off the blockchain? Insights on off-chaining computation and data, in: Proc. European Conf. Service-Oriented and Cloud Computing (ESOCC), Springer. doi:10.1007/978-3-319-67262-5_1.

Eren, H., Karaduman, O., Gençoğlu, M., 2025. Security challenges and performance trade-offs in on-chain and off-chain blockchain storage: a comprehensive review. Appl. Sci. 15, 3225. doi:10.3390/app15063225.

Ethereum Foundation, 2025. Ethereum: open source blockchain for decentralized applications. https://ethereum.org/en/. Accessed: July 2025.

Frikha, T., Chaabane, F., Aouinti, N., Cheikhrouhou, O., Ben Amor, N., Kerrouche, A., 2021. Implementation of blockchain consensus algorithm on embedded architecture. J. Comput. Netw. Commun. 2021, 9918697. doi:10.1155/2021/9918697.

Frikha, T., Chaabane, F., Halima, R., et al., 2023. Embedded decision support platform based on multi-agent systems. Multimed. Tools Appl. 82, 32607–32633. doi:10.1007/s11042-023-14843-x.

Gao, Y., Pan, Q., Liu, Y., Lin, H., Chen, Y., Wen, Q., 2021. The notarial office in e-government: a blockchain-based solution. IEEE Access 9, 44411–44425. doi:10.1109/ACCESS.2021.3066184.

Ghadi, Y., Mazhar, T., Shahzad, T., et al., 2024. The role of blockchain to secure internet of medical things. Sci. Rep. 14, 18422. doi:10.1038/s41598-024-68529-x.

Gong, F., Min, X., 2023. An overview of blockchain scalability for storage, in: Proc. 26th Int. Conf. Computer Supported Cooperative Work in Design (CSCWD), IEEE. pp. 516–521. doi:10.1109/CSCWD57460.2023.10152720.

Guo, H., Yu, X., 2022. A survey on blockchain technology and its security. Blockchain Res. Appl. 3, 100067. doi:10.1016/j.bcra.2022.100067.

Horng, J., Chang, C., Li, G., Lee, W., Hwang, S., 2021. Blockchain-based reversible data hiding for securing medical images. Secur. Commun. Netw. 2021, 9943402. doi:10.1155/2021/9943402.

Karaman, B., et al., 2024. Assessing the significance of longitudinal data in Alzheimer's disease forecasting. arXiv:2405.17352.

Karmani, M., et al., 2009. Efficient hardware architecture of 2D-scan-based wavelet watermarking for image and video. Comput. Stand. Interfaces 31, 801–811. doi:10.1016/j.csi.2008.09.015.

Kostopoulos, N., Stamatiou, Y., Halkiopoulos, C., Antonopoulou, H., 2025. Blockchain applications in the military domain: a systematic review. Technologies 13, 23. doi:10.3390/technologies13010023.

Kotter, E., Marti-Bonmati, L., Brady, A., Desouza, N., 2021. ESR white paper: blockchain and medical imaging. Insights Imaging 12, 87. doi:10.1186/s13244-021-01029-y.

Ktari, J., Affes, N., Frikha, T., Hamdi, M., Hamam, H., 2024a. Embedded low power blockchain traceability solution for university classroom attendance. Jordan J. Electr. Eng. 10, 465–483. doi:10.5455/jjee.204-1703426260.

Ktari, J., Frikha, T., Ben Amor, N., Louraidh, L., Elmannai, H., Hamdi, M., 2022. IoMT-based platform for e-health monitoring based on the blockchain. Electronics 11, 2314. doi:10.3390/electronics11152314.

Ktari, J., Frikha, T., Hamdi, M., Hamam, H., 2024b. Enhancing blockchain consensus with FPGA: accelerating implementation for efficiency. IEEE Access 12, 44773–44785. doi:10.1109/ACCESS.2024.3379374.

Kudva, S., Badsha, S., Sengupta, S., Khalil, I., Zomaya, A., 2021. Towards secure and practical consensus for blockchain-based VANET. Inf. Sci. 545, 170–187. doi:10.1016/j.ins.2020.07.060.

Kumar, A., et al., 2024. Ensemble activation enabled deep classifier for Alzheimer's disease detection in the blockchain-enabled distributed edge environment. Int. J. Inf. Technol. doi:10.1007/s41870-024-01833-x.

Kyriazis, N., 2021. A survey on volatility fluctuations in the decentralized cryptocurrency financial assets. J. Risk Financ. Manag. 14, 293.

Liu, Q., Peng, Y., Xu, M., Jiang, H., Wu, J., Wang, G., 2024. MPV: enabling fine-grained query authentication in hybrid-storage blockchain. IEEE Transactions on Knowledge and Data Engineering doi:10.1109/TKDE.2024.3359173.

Lunardi, R., Alharby, M., Nunes, H., Zorzo, A., Dong, C., Middleton, M., 2020. Context-based consensus for appendable-block blockchains, in: Proceedings of the 2020 IEEE International Conference on Blockchain, IEEE. pp. 401–408. doi:10.1109/Blockchain50366.2020.00058.

Nakamoto, S., 2008. Bitcoin: a peer-to-peer electronic cash system. https://www.bitcoin.org/bitcoin.pdf.

Nguyen, D., Ding, M., Pathirana, P., Seneviratne, A., 2021. Blockchain-enabled Internet of Medical Things to combat COVID-19. IEEE Internet Things Mag. 4, 44–49. doi:10.1109/IOTM.0001.2100020.

NVIDIA Corporation, 2025. NVIDIA Jetson Nano Developer Kit. https://developer.nvidia.com/embedded/jetson-nano-developer-kit. Accessed: August 2025.

Ren, W., Wan, X., Gan, P., 2021. A double-blockchain solution for agricultural sampled data security in Internet of Things network. Future Gener. Comput. Syst. 117, 453–461.

Roumeliotis, C., Dasygenis, M., Lazaridis, V., Dossis, M., 2024. Blockchain and digital twins in smart Industry 4.0: the use case of supply chain—a review of integration techniques and applications. Designs 8, 105. doi:10.3390/designs8060105.

Sakakibara, Y., Nakamura, K., Matsutani, H., 2017. An FPGA NIC based hardware caching for blockchain, in: Proc. 8th Int. Symp. Highly Efficient Accelerators and Reconfigurable Technologies (HEART), ACM. pp. 1–6. doi:10.1145/3120895.3120897.

Sel, D., Jacobsen, H., et al., 2018. Towards solving the data availability problem for sharded Ethereum, in: Proc. 2nd Workshop on Scalable and Resilient Infrastructures for Distributed Ledgers, ACM. doi:10.1145/3284764.3284769.

Singh, B., Tatiya, M., Shrivastava, A., Verma, D., Srivastava, A., Rana, A., 2022. Detection of Alzheimer's disease using deep learning, blockchain, and IoT cognitive data, in: Proceedings of the 2022 International Conference on Information and Communication Technology for Competitive Strategies, IEEE. pp. 863–869. doi:10.1109/ICTACS56270.2022.9988058.

Talha, A., Frikha, T., Ktari, J., Hamam, H., 2023. Revolutionizing Tunisian agricultural traceability with blockchain: exploring Aries and Ethereum solutions. J. Inf. Assur. Secur. 20.

Tiwari, N., Ranjan, P., Biswal, P., et al., 2025. Survey and analysis of blockchain technologies with respect to: properties, algorithms, architecture, models, evolution and framework. Multimed. Tools Appl. 84, 13571–13615. doi:10.1007/s11042-024-19580-3.

Wang, X., Zhang, Z., Yang, X., Wang, G., Zhang, A., Yuan, G., 2022. Efficient authentication processing for spatial keyword queries in hybrid storage blockchain, in: Proc. 4th Int. Conf. Blockchain Technology (ICBCT), ACM. pp. 22–30. doi:10.1145/3532640.3532644.

Wang, Z., Yu, L., Zhou, L., 2025. Navigating the blockchain-driven transformation in Industry 4.0: opportunities and challenges for economic and management innovations. J. Knowl. Econ. 16, 3507–3549. doi:10.1007/s13132-024-02007-7.

Wöhrer, M., Zdun, U., 2021. Architectural design decisions for blockchain-based applications, in: Proc. IEEE Int. Conf. Blockchain and Cryptocurrency (ICBC), IEEE. pp. 1–5. doi:10.1109/ICBC51069.2021.9461109.

Xu, J., Wang, C., Jia, X., 2023. A survey of blockchain consensus protocols. ACM Comput. Surv. 55, 278. doi:10.1145/3579845.

Zhang, X., et al., 2026. Securing federated learning with blockchain in the medical field: systematic literature review. J. Med. Internet Res. 28, e79052. doi:10.2196/79052.

Zhou, L., Diro, A., Saini, A., Kaisar, S., Hiep, P., 2024. Leveraging zero knowledge proofs for blockchain-based identity sharing: a survey of advancements, challenges and opportunities. J. Inf. Secur. Appl. 80, 103678. doi:10.1016/j.jisa.2023.103678.